Faculty of Computer Science and Mathematics
Universiti Malaysia Terengganu

LAB REPORT

CSE3023-PEMBANGUNAN APLIKASI BERASASKAN WEB

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Lab Report	2
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Exercise Lab 2:

add_product.html

```html
<html>
    <head>
        <title>Add Product</title>
    </head>
    <body>
        <h2>Add Product</h2>
        <form action="insert" method="post">

            Name:
            <input type="text" name="name"><br><br>

            Category:
            <input type="text" name="category"><br><br>

            Price:
            <input type="text" name="price"><br><br>

            Quantity:
            <input type="text" name="quantity"><br><br>

            <input type="submit" value="Add Product">
        </form>
        <br>
        <a href="view">View Products</a>
    </body>
</html>
```

DBConnection.java

```java
package com.lab.dao;
import java.sql.Connection;
import java.sql.DriverManager;
/**
 *
 * @author AIZAT
 */
public class DBConnection {

    private static String url = "jdbc:mysql://localhost:3306/cse3023";
    private static String username = "root";
    private static String password = "";

    public static Connection getConnection() {
        Connection con = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            con = DriverManager.getConnection(
                    url, username, password);
        } catch (Exception e) {
            e.printStackTrace();
        }
        return con;
    }
}
```

ProductDAO.java

```java
package com.lab.dao;
import java.sql.*;
import java.util.ArrayList;
import java.util.List;
import com.lab.model.Product;
/**
 *
 * @author AIZAT
 */
public class ProductDAO {
    // INSERT
    public boolean insertProduct(Product p) {

        boolean status = false;

        try {

            Connection con =
                DBConnection.getConnection();

            String sql =
                "INSERT INTO products(name,category,price,quantity) VALUES(?,?,?,?)";

            PreparedStatement ps =
                con.prepareStatement(sql);

            ps.setString(1, p.getName());
            ps.setString(2, p.getCategory());
            ps.setDouble(3, p.getPrice());
            ps.setInt(4, p.getQuantity());

            status = ps.executeUpdate() > 0;

            con.close();

        } catch (Exception e) {
            e.printStackTrace();
        }

        return status;
    }
    // SELECT ALL
    public List<Product> getAllProducts() {
        List<Product> list =
            new ArrayList<>();
        try {
            Connection con =
                DBConnection.getConnection();
            String sql =
                "SELECT * FROM products";
```

ProductDAO.java

```java
                "SELECT * FROM products";
            PreparedStatement ps =
                con.prepareStatement(sql);
            ResultSet rs =
                ps.executeQuery();
            while (rs.next()) {
                Product p = new Product();
                p.setId(rs.getInt("id"));
                p.setName(rs.getString("name"));
                p.setCategory(rs.getString("category"));
                p.setPrice(rs.getDouble("price"));
                p.setQuantity(rs.getInt("quantity"));
                list.add(p);
            }
            con.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
        return list;
    }
    // DELETE
    public boolean deleteProduct(int id) {
        boolean status = false;
        try {
            Connection con =
                DBConnection.getConnection();

            String sql =
                "DELETE FROM products WHERE id=?";

            PreparedStatement ps =
                con.prepareStatement(sql);
            ps.setInt(1, id);
            status = ps.executeUpdate() > 0;
            con.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
        return status;
    }
    // UPDATE
    public boolean updateProduct(Product p) {
        boolean status = false;
        try {
            Connection con =
                DBConnection.getConnection();

            String sql =
                "UPDATE products SET name=?,category=?,price=?,quantity=? WHERE id=?";
```

```java
                "UPDATE products SET name=?,category=?,price=?,quantity=? WHERE id=?";

            PreparedStatement ps =
                con.prepareStatement(sql);
            ps.setString(1, p.getName());
            ps.setString(2, p.getCategory());
            ps.setDouble(3, p.getPrice());
            ps.setInt(4, p.getQuantity());
            ps.setInt(5, p.getId());
            status = ps.executeUpdate() > 0;
            con.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
        return status;
    }
}
```

DeleteProductServlet.java

```java
package servlet;
import com.lab.dao.ProductDAO;

import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
/**
 *
 * @author AIZAT
 */

public class DeleteProductServlet extends HttpServlet {

    protected void doGet(
            HttpServletRequest request,
            HttpServletResponse response)
            throws ServletException, IOException {

        int id =
          Integer.parseInt(
            request.getParameter("id"));

        ProductDAO dao =
            new ProductDAO();

        dao.deleteProduct(id);

        response.sendRedirect("view");
    }
}
```

EditProductServlet.java

```java
public class EditProductServlet extends HttpServlet {

    protected void doGet(
            HttpServletRequest request,
            HttpServletResponse response)
            throws ServletException, IOException {

        int id =
            Integer.parseInt(
            request.getParameter("id"));

        ProductDAO dao =
            new ProductDAO();

        List<Product> list =
            dao.getAllProducts();

        Product product = null;

        for (Product p : list) {
            if (p.getId() == id) {
                product = p;
                break;
            }
        }
        response.setContentType("text/html");
        PrintWriter out =
            response.getWriter();
        out.println("<h2>Edit Product</h2>");
        out.println("<form action='update' method='post'>");
        out.println("<input type='hidden' name='id' value='"+product.getId()+"'>");
        out.println("Name:");
        out.println("<input type='text' name='name' value='"+product.getName()+"'><br><br>");
        out.println("Category:");
        out.println("<input type='text' name='category' value='"+product.getCategory()+"'><br><br>");
        out.println("Price:");
        out.println("<input type='text' name='price' value='"+product.getPrice()+"'><br><br>");
        out.println("Quantity:");
        out.println("<input type='text' name='quantity' value='"+product.getQuantity()+"'><br><br>");
        out.println("<input type='submit' value='Update Product'>");
        out.println("</form>");
    }
}
```

UpdateProductServlet.java

```java
public class UpdateProductServlet extends HttpServlet {

    protected void doPost(
            HttpServletRequest request,
            HttpServletResponse response)
            throws ServletException, IOException {
        int id =
            Integer.parseInt(
            request.getParameter("id"));
        String name =
            request.getParameter("name");
        String category =
            request.getParameter("category");
        double price =
            Double.parseDouble(
            request.getParameter("price"));

        int quantity =
            Integer.parseInt(
            request.getParameter("quantity"));
        Product p = new Product();

        p.setId(id);
        p.setName(name);
        p.setCategory(category);
        p.setPrice(price);
        p.setQuantity(quantity);

        ProductDAO dao =
            new ProductDAO();

        dao.updateProduct(p);

        response.sendRedirect("view");
    }
}
```

ViewProductServlet.java

```java
 * @author AIZAT
 */
public class ViewProductServlet extends HttpServlet {

    protected void doGet(
            HttpServletRequest request,
            HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out =
            response.getWriter();

        ProductDAO dao =
            new ProductDAO();

        List<Product> list =
            dao.getAllProducts();

        out.println("<h2>Product List</h2>");
        out.println("<table border='1'>");
        out.println(
          "<tr><th>ID</th><th>Name</th><th>Category</th>"
          + "<th>Price</th><th>Quantity</th>"
          + "<th>Edit</th><th>Delete</th></tr>");

        for (Product p : list) {

            out.println("<tr>");
            out.println("<td>"+p.getId()+"</td>");
            out.println("<td>"+p.getName()+"</td>");
            out.println("<td>"+p.getCategory()+"</td>");
            out.println("<td>"+p.getPrice()+"</td>");
            out.println("<td>"+p.getQuantity()+"</td>");
            out.println(
             "<td><a href='edit?id="+p.getId()+"'>Edit</a></td>");

            out.println(
             "<td><a href='delete?id="+p.getId()+"'>Delete</a></td>");

            out.println("</tr>");
        }
        out.println("</table>");
    }
}
```

Output:

Add Product

localhost:8080/ProductInventorySystem/add_product.html

Add Product

Name: Solar

Category: lampu

Price: 1054

Quantity: 34

Add Product

View Products

localhost:8080/ProductInventoryS

localhost:8080/ProductInventorySystem/view

Product List

ID	Name	Category	Price	Quantity	Edit	Delete
1	Wireless Mouse	Electronics	25.99	15	Edit	Delete
2	Mechanical Keyboard	Electronics	89.0	20	Edit	Delete
4	Pistol	gun	1055.0	5	Edit	Delete
5	Gam gajah	gula gula	500.0	50	Edit	Delete
6	Solar	lampu	1054.0	34	Edit	Delete